MuSR-Stress: A Benchmark for Long-Context and Revisable Belief Reasoning

James Cheng
Stanford University
james@example.edu

March 1, 2026

Abstract

We present **MuSR-Stress**, a benchmark for evaluating reasoning systems along two axes: long-context reasoning and dynamic belief consistency under streaming evidence and counterfactual correction. The benchmark includes two tracks (`long_context` and `dynamic_belief`) with standardized train/dev/test splits. We report metrics beyond final-answer accuracy, including update consistency, calibration (Brier score), required belief revision, recovery rate, and recovery latency. Baseline results show high trajectory consistency but weaker revision success, suggesting that counterfactual corrections remain challenging for current models.

1 Motivation

State-of-the-art language models perform strongly on static QA, but many real settings require reasoning over evolving evidence. Models should maintain coherent beliefs and revise them when new evidence invalidates prior conclusions.

MuSR-Stress targets this gap by jointly evaluating:

- long-context integration under distractor noise,
- dynamic belief revision under streamed and counterfactual evidence.

2 Background & Related Work

MuSR foundation. MuSR introduced structured multi-step soft reasoning with logic-tree-based supervision and narrative contexts. MuSR-Stress extends this setup from static one-shot QA to trajectory-level belief tracking and revision.

Tracks in the current benchmark. The current benchmark provides two tracks:

- `long_context`: stresses long-horizon context integration by injecting distractor cases around the target narrative.
- `dynamic_belief`: unified dynamic track mixing stream-only and counterfactual cases, testing sequential belief updates and correction.

Current scale. Each track has 250 examples split into 175 train / 37 dev / 38 test.

Dynamic-belief format. Each instance contains a question, candidate suspects, gold answer, ordered evidence rounds, and metadata fields such as `has_counterfactual`, `flip_required`, and correction onset round.

3 Proposed Method & Experiments

3.1 Current Pipeline

Seed Sampling (Madlib pools) → Per-Suspect Logic Tree Construction → Validator-Gated Recursive Completion → Murderer/Innocent Chapter Generation → Story Assembly + QA Instance → Stressor Builders (long_context, dynamic_belief) → Train/Dev/Test Split + Evaluation

Figure 1: Current MuSR-Stress pipeline implemented in this repository.

3.2 Generation Settings

The dynamic track is generated with configurable round limits and correction rates. Current v2 settings:

- `max_rounds`=40
- `counterfactual_rate`=0.7
- `flip_rate`=0.7
- `evidence_source`=narrative

Evidence rounds are narrative sentences split from the case's story prose (not internal logic-tree facts), so the model reasons from the same natural-language narrative a human reader would see.

This produces both:

- **stream-only** cases (no contradiction correction), and
- **counterfactual** cases (with a correction block that may require answer revision).

3.3 Example: Counterfactual Case

Table 1 shows a representative `flip_required` counterfactual instance (`dynamic_belief_187_q0`) from the test split. Suspects are Dale and Letti; the victim is Josephine. Early narrative rounds describe Dale's confrontation with Josephine, his suspicious multiple driver's licenses, his frequent presence at her café, and an invitation to her house on the day of the murder — all pointing toward Dale. At round 21 the counterfactual block withdraws the witness timeline against Letti and introduces new physical evidence (phone location, weapon purchase, threatening messages) implicating Dale. Post-correction narrative continues, requiring the model to hold the revised belief through the remaining rounds.

3.4 Prompt Format

At each round t the model receives a single prompt containing the question, the suspect list, and all evidence accumulated so far (rounds 1 through t), then returns a JSON object with a `top_suspect` field and a `scores` dict (one entry per suspect). Scores are normalized to a probability distribution for metric computation. Figure 2 shows the prompt template.

3.5 Evaluation Protocol

Let $T = (r_1, r_2, \ldots, r_n)$ be the sequence of rounds for a case, $\hat{s}_t$ the model's top suspect at round t, $p_t(s)$ the model's probability for suspect s at round t, s^* the gold suspect, and r_{cf} the counterfactual onset round (where applicable). For counterfactual cases, the *active gold* switches at r_{cf}: $g_t = s^*_{\text{before}}$ if $t < r_{cf}$, else $g_t = s^*_{\text{after}}$.

Final accuracy.

$$\text{final\_accuracy} = \mathbf{1}[\hat{s}_n = s^*]$$

Table 1: Example counterfactual case (`dynamic_belief_187_q0`). Gold switches from Letti → Dale at round 21.

Round	Type	Evidence
1	narrative	Detective Winston investigates Josephine's death; prime suspects Dale and Letti are under the microscope.
4	narrative	A witness statement details how Dale angrily confronted Josephine after discovering her new relationship.
6	narrative	Reports include Josephine's new lover, the confrontation with Dale, suspicious multiple driver's licenses found at Dale's house, and Dale's frequent presence at the victim's café.
8	narrative	Josephine herself handed Dale an invitation to her house on the day of the murder, when no one else was home.
19	narrative	Winston noticed Dale hastily shove a few driver's licenses into his wallet.
21	**counterfactual**	**Correction: the key witness timeline against Letti had an incorrect timestamp and is now withdrawn.**
22	**counterfactual**	**Follow-up records place Dale's phone near the scene during the murder window, show a matching weapon purchase, and include messages documenting escalating conflict with Josephine.**
23+	narrative	Story continues — model must maintain revised belief (Dale) through rounds 23–40.

Brier score.

$$\mathrm{BS}_t = \sum_{s \in \mathcal{S}} \left(p_t(s) - \mathbf{1}[s = g_t]\right)^2$$

brier_final $= \mathrm{BS}_n$; **brier_mean** $= \frac{1}{n}\sum_{t=1}^{n} \mathrm{BS}_t$.

Update consistency. Let $\mathcal{F}$ be the set of round indices where $\hat{s}_t \neq \hat{s}_{t-1}$. In flip-required counterfactual cases, the first transition in $\mathcal{F}$ with $t \geq r_{cf}$ and $\hat{s}_t = s^*_{\text{after}}$ is exempted (it is a correct revision). The flip penalty is:

$$\text{flip\_penalty} = \frac{|\mathcal{F}_{\text{penalized}}|}{n-1}$$

For the trajectory penalty, let τ be the subsequence of rounds used (all rounds for stream-only; post-r_{cf} rounds for counterfactual cases):

$$\text{traj\_penalty} = \frac{\sum_{t \in \tau,\, t>1} \max(0,\ p_{t-1}(\hat{s}_n) - p_t(\hat{s}_n))}{|\tau| - 1}$$

$$\text{update\_consistency} = \max\Big(0,\ \min\Big(1,\ 1 - \tfrac{1}{2}(\text{flip\_penalty} + \text{traj\_penalty})\Big)\Big)$$

Flip when required. Defined only for flip-required counterfactual cases. Let pre- and post-correction predictions be $\hat{s}_{\text{pre}} = \hat{s}_{r_{cf}-1}$ and $\hat{s}_{\text{post}} = \hat{s}_n$:

$$\text{flip\_when\_required} = \mathbf{1}[\hat{s}_{\text{post}} = s^*_{\text{after}}]$$

Stability when not required. Defined only for counterfactual cases where `flip_required=False`:

$$\text{stability\_when\_not\_required} = \mathbf{1}[\hat{s}_{\text{pre}} = \hat{s}_{\text{post}}]$$

```
You are evaluating a murder mystery as evidence arrives.

Question: Who is the most likely murderer?
Suspects:
- Dale
- Letti

Evidence so far:
1. [round 1 evidence text]
2. [round 2 evidence text]
...
k. [round k evidence text]

Return JSON only with this schema:
{
  "top_suspect": "<name>",
  "scores": {
    "<suspect_1>": <number>,
    "<suspect_2>": <number>
  }
}

Rules:
- Include every suspect exactly once in "scores".
- Scores can be any non-negative numbers.
- "top_suspect" must be one of the suspects.
```

Figure 2: Prompt sent to the model at each round t. Evidence is accumulated incrementally; the model re-evaluates after every new sentence.

Recovery rate and latency. Defined only for flip-required counterfactual cases. Let $t^* = \min\{t \geq r_{cf} : \hat{s}_t = s^*_{\text{after}}\}$ (undefined if never reached):

$$\text{recovery\_rate} = \mathbf{1}[t^* \text{ exists}] \qquad \text{recovery\_latency} = t^* - r_{cf}$$

Recovery latency is reported only over recovered cases (where t^* exists), so a value of 0 indicates immediate correction at the counterfactual round.

3.6 Preliminary Baseline Results

We evaluate instruct models at temperature 0 on the `dynamic_belief` test split ($n = 38$).

Table 2: Dynamic-belief test results ($n = 38$).

Metric	Qwen2.5-7B-Instruct	Llama-3.3-70B-Instruct
Final accuracy	0.789	TBD
counterfactual subset (n=28)	0.893	TBD
stream-only subset (n=10)	0.500	TBD
Update consistency	0.940	TBD
Brier final	0.344	TBD
Brier mean	0.431	TBD
Flip when required	**0.952**	TBD
Stability when not required	1.000	TBD
Recovery rate	**1.000**	TBD
Recovery latency (rounds)	0.43	TBD

Key observations. Qwen2.5-7B-Instruct handles counterfactual revision remarkably well: `flip_when_required`=0.952, `recovery_rate`=1.0, and `recovery_latency`=0.43 rounds. This suggests the model responds quickly and reliably to explicit correction signals. The more surprising finding is the gap between counterfactual accuracy (0.893) and stream-only accuracy (0.500): cases *without* a correction are harder than cases *with* one. A likely explanation is that counterfactual cases provide an anchoring signal — the correction text explicitly names the revised suspect — whereas stream-only cases require reasoning purely from accumulated evidence. Note that the stream-only subset is small ($n = 10$), so this gap should be interpreted cautiously.

3.7 Dataset Composition

The full `dynamic_belief` track (250 cases, 70/15/15 train/dev/test split, seed=7) breaks down as follows from the generation manifest:

Table 3: MuSR-Stress dynamic_belief track composition.

Split	Total	CF / Stream-only
Train	175	—
Dev	37	—
Test	38	—
Total	250	170 CF / 80 stream-only
flip-required CF	120	(of 170 CF)
flip-not-required CF	50	(of 170 CF)

4 Next Steps

Current limitations include single-domain emphasis and single-seed headline reporting. Next steps are:

- multi-seed reporting with mean/variance,
- failure taxonomy (spurious flip vs missed required flip),
- evaluation of additional models on both tracks (`long_context`, `dynamic_belief`),
- a combined track stressing both long-context attention and dynamic belief revision simultaneously,
- extension of dynamic-belief generation to other domains beyond murder mystery.

5 Conclusion

MuSR-Stress provides a practical benchmark for evaluating not only correctness, but also whether models maintain and revise beliefs appropriately as evidence evolves across two complementary tracks. Results motivate targeted progress on non-monotonic belief updating and long-context attention under distractor noise.